SG No. 172 C_6^5 $P6_4$ [hexagonal]

* plus set: $+[0,\ 0,\ 0]$

* Wyckoff site: `3a`, site symmetry: `2..`

Table 1: Wyckoff bond: `3a@3a`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[X,\ Y,\ 0]$	$[0,\ 0,\ z]$	`[1,-4]`
2	$[-Y,\ X-Y,\ 0]$	$\left[0,\ 0,\ z+\frac{1}{3}\right]$	`[2,-5]`
3	$[-X+Y,\ -X,\ 0]$	$\left[0,\ 0,\ z+\frac{2}{3}\right]$	`[3,-6]`

Table 2: Wyckoff bond: `3b@3a`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[0,\ 0,\ Z]$	$[0,\ 0,\ z]$	`[1,4]`
2	$[0,\ 0,\ Z]$	$\left[0,\ 0,\ z+\frac{1}{3}\right]$	`[2,5]`
3	$[0,\ 0,\ Z]$	$\left[0,\ 0,\ z+\frac{2}{3}\right]$	`[3,6]`

Table 3: Wyckoff bond: `6c@3a`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X,\ Y,\ Z]$	$[0,\ 0,\ z]$	`[1]`
2	$[-Y,\ X-Y,\ Z]$	$\left[0,\ 0,\ z+\frac{1}{3}\right]$	`[2]`
3	$[-X+Y,\ -X,\ Z]$	$\left[0,\ 0,\ z+\frac{2}{3}\right]$	`[3]`
4	$[-X,\ -Y,\ Z]$	$[0,\ 0,\ z]$	`[4]`
5	$[Y,\ -X+Y,\ Z]$	$\left[0,\ 0,\ z+\frac{1}{3}\right]$	`[5]`
6	$[X-Y,\ X,\ Z]$	$\left[0,\ 0,\ z+\frac{2}{3}\right]$	`[6]`

* Wyckoff site: `3b`, site symmetry: `2..`

Table 4: Wyckoff bond: `3a@3b`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[X,\ Y,\ 0]$	$\left[\frac{1}{2},\ \frac{1}{2},\ z\right]$	`[1,-4]`
2	$[-Y,\ X-Y,\ 0]$	$\left[\frac{1}{2},\ 0,\ z+\frac{1}{3}\right]$	`[2,-5]`
3	$[-X+Y,\ -X,\ 0]$	$\left[0,\ \frac{1}{2},\ z+\frac{2}{3}\right]$	`[3,-6]`

Table 5: Wyckoff bond: `3b@3b`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[0,\ 0,\ Z]$	$\left[\frac{1}{2},\ \frac{1}{2},\ z\right]$	`[1,4]`
2	$[0,\ 0,\ Z]$	$\left[\frac{1}{2},\ 0,\ z+\frac{1}{3}\right]$	`[2,5]`
3	$[0,\ 0,\ Z]$	$\left[0,\ \frac{1}{2},\ z+\frac{2}{3}\right]$	`[3,6]`

Table 6: Wyckoff bond: `6c@3b`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, z\right]$	`[1]`
2	$[-Y, X-Y, Z]$	$\left[\frac{1}{2}, 0, z+\frac{1}{3}\right]$	`[2]`
3	$[-X+Y, -X, Z]$	$\left[0, \frac{1}{2}, z+\frac{2}{3}\right]$	`[3]`
4	$[-X, -Y, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, z\right]$	`[4]`
5	$[Y, -X+Y, Z]$	$\left[\frac{1}{2}, 0, z+\frac{1}{3}\right]$	`[5]`
6	$[X-Y, X, Z]$	$\left[0, \frac{1}{2}, z+\frac{2}{3}\right]$	`[6]`

* Wyckoff site: `6c`, site symmetry: `1`

Table 7: Wyckoff bond: `6a@6c`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[x, y, z]$	`[1]`
2	$[-Y, X-Y, Z]$	$\left[-y, x-y, z+\frac{1}{3}\right]$	`[2]`
3	$[-X+Y, -X, Z]$	$\left[-x+y, -x, z+\frac{2}{3}\right]$	`[3]`
4	$[-X, -Y, Z]$	$[-x, -y, z]$	`[4]`
5	$[Y, -X+Y, Z]$	$\left[y, -x+y, z+\frac{1}{3}\right]$	`[5]`
6	$[X-Y, X, Z]$	$\left[x-y, x, z+\frac{2}{3}\right]$	`[6]`